Day 012 | Unit 01 | Teacher plan

Positive Rational Numbers: Decimals and Division

Interpret Whole-Number Remainders

Standards: 6.NS.B.2

Learning target: I can divide multi-digit whole numbers and check the quotient.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 12

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01 / Learn

Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder.

Worked example 1

Use long division: 546 ÷ 7. Show the algorithm and check.

1. Estimate how many digits the quotient should have. Work from the leftmost digits of the dividend.

2. 54 ÷ 7: write 7, subtract 49, remainder 5

3. bring down the next digit: 56 ÷ 7: write 8, subtract 56, remainder 0.

4. Check: 7 × 78 = 546.

Answer: 78

Worked example 2

Use long division: 11,312 ÷ 33. Show the algorithm and check.

1. Estimate the quotient first.

2. 113 ÷ 33: write 3, subtract 99, remainder 14

3. bring down the next digit: 141 ÷ 33: write 4, subtract 132, remainder 9

4. bring down the next digit: 92 ÷ 33: write 2, subtract 66, remainder 26.

5. Check: 33 × 342 + 26 = 11312.

Answer: 342 R 26

Day 012 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use long division: 450 ÷ 6. Show the algorithm and check.

02 / Constructed response

Use long division: 448 ÷ 8. Show the algorithm and check.

03 / Constructed response

Use long division: 440 ÷ 5. Show the algorithm and check.

04 / Constructed response

Use long division: 6,647 ÷ 23. Show the algorithm and check.

05 / Constructed response

Use long division: 6,325 ÷ 23. Show the algorithm and check.

06 / Constructed response

Use long division: 2,327 ÷ 13. Show the algorithm and check.

07 / Constructed response

Use long division: 11,533 ÷ 30. Show the algorithm and check.

08 / Model and constructed response

Use long division: 13,555 ÷ 36. Show the algorithm and check.

09 / Model and constructed response

Use long division: 4,262 ÷ 34. Show the algorithm and check.

10 / Constructed response

A club has 6605 cards. Each full packet holds 25. How many full packets can it make, and how many cards remain?

Day 012 | Unit 01 | Answer key

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1. 75

45 ÷ 6: write 7, subtract 42, remainder 3; bring down the next digit: 30 ÷ 6: write 5, subtract 30, remainder 0. Check: 6 × 75 = 450.

6.NS.B.2 | Constructed response

2. 56

44 ÷ 8: write 5, subtract 40, remainder 4; bring down the next digit: 48 ÷ 8: write 6, subtract 48, remainder 0. Check: 8 × 56 = 448.

6.NS.B.2 | Constructed response

3. 88

44 ÷ 5: write 8, subtract 40, remainder 4; bring down the next digit: 40 ÷ 5: write 8, subtract 40, remainder 0. Check: 5 × 88 = 440.

6.NS.B.2 | Constructed response

4. 289

66 ÷ 23: write 2, subtract 46, remainder 20; bring down the next digit: 204 ÷ 23: write 8, subtract 184, remainder 20; bring down the next digit: 207 ÷ 23: write 9, subtract 207, remainder 0. Check: 23 × 289 = 6647.

6.NS.B.2 | Constructed response

5. 275

63 ÷ 23: write 2, subtract 46, remainder 17; bring down the next digit: 172 ÷ 23: write 7, subtract 161, remainder 11; bring down the next digit: 115 ÷ 23: write 5, subtract 115, remainder 0. Check: 23 × 275 = 6325.

6.NS.B.2 | Constructed response

6. 179

23 ÷ 13: write 1, subtract 13, remainder 10; bring down the next digit: 102 ÷ 13: write 7, subtract 91, remainder 11; bring down the next digit: 117 ÷ 13: write 9, subtract 117, remainder 0. Check: 13 × 179 = 2327.

6.NS.B.2 | Constructed response

7. 384 R 13

115 ÷ 30: write 3, subtract 90, remainder 25; bring down the next digit: 253 ÷ 30: write 8, subtract 240, remainder 13; bring down the next digit: 133 ÷ 30: write 4, subtract 120, remainder 13. Check: 30 × 384 + 13 = 11533.

6.NS.B.2 | Constructed response

8. 376 R 19

135 ÷ 36: write 3, subtract 108, remainder 27; bring down the next digit: 275 ÷ 36: write 7, subtract 252, remainder 23; bring down the next digit: 235 ÷ 36: write 6, subtract 216, remainder 19. Check: 36 × 376 + 19 = 13555.

6.NS.B.2 | Model and constructed response

9. 125 R 12

42 ÷ 34: write 1, subtract 34, remainder 8; bring down the next digit: 86 ÷ 34: write 2, subtract 68, remainder 18; bring down the next digit: 182 ÷ 34: write 5, subtract 170, remainder 12. Check: 34 × 125 + 12 = 4262.

6.NS.B.2 | Model and constructed response

10. 264 full packets; 5 cards remain

6605 = 25 × 264 + 5.

6.NS.B.2 | Constructed response